

Universal Balance Laws: Two Compressed Principles

A Concise Statement of UBL

Dustin

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INTRODUCTION

Complex systems — whether economic markets, physical networks, or social graphs — often conceal simple update rules beneath their apparent disorder. Universal Balance Laws (UBL) is a minimal framework that proposes two such rules: one governing how quantities flow and accumulate between agents, and one governing how influence attenuates with distance. Together they constitute a compressed description of local growth and global reach.

LAW 1 — THE FLOW LAW

The Flow Law governs exchange and accumulation across any step of a process. Let D represent a demand quantity, S a supply or scaling factor, and D_{flow} the amount transiting in a given step. The law is stated as two coupled updates:

$$D_{\text{next}} = D + S \times D_{\text{flow}}$$

$$S_{\text{next}} = S + k \times S \times D_{\text{flow}}$$

At each step, the demand quantity D grows by exactly as much as the scaling factor S amplifies the incoming flow. Simultaneously, S itself increases in proportion to $k \times S$

$\times D_{\text{flow}}$ — capturing the principle that a larger supply base is better positioned to grow further, compounding with each exchange. The coupling constant k is a small positive scalar that calibrates how strongly the supply side responds to each unit of flow; varying k interpolates between sluggish and explosive accumulation regimes.

LAW 2 — THE DISTANCE LAW

The Distance Law governs how influence, gravity, or signal strength decays as separation grows. Let G represent the effective influence between two entities — nodes, agents, or physical objects. Across each unit of distance $dist$, that influence obeys:

$$G_{\text{next}} = G / dist^2$$

The quantity G may represent gravitational attraction, information fidelity, economic coupling, or any analogous measure of reach. Dividing by the square of the distance is the defining operation: at twice the separation, only one quarter of the original influence remains; at three times the distance, only one ninth persists. This inverse-square geometry imposes a natural locality on all interaction, ensuring that the strong coupling described by Law 1 operates predominantly at short range.

CONCLUSION

Taken together, the two laws sketch a coherent picture: quantities flow and self-amplify locally under Law 1, while Law 2 ensures that no influence propagates without geometric cost. The compression of these dynamics into a pair of scalar update rules — requiring only arithmetic to evaluate — is the central claim of UBL. Whether the framework generalizes faithfully to empirical systems remains an open and tractable question; its value at this stage lies in providing a shared, minimal language for reasoning about flow and reach.